



Influence of tip clearance on heat transfer efficiency in micro-cylinders-group heat sink

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ABSTRACT

De-ionized water, used as working fluid, flows through the in-line and staggered array micro-cylinders-group heat sinks with different tip clearances, which are made up of micro-cylinders with hydraulic diameters of 0.5 mm and heights of 0.5 mm. The resistance characteristic and the Nusselt numbers in micro-cylinders-group heat sinks at in-line array and staggered array can be experimentally obtained over Reynolds number ranging from 8 to 400, so the heat transfer efficiency in the micro-cylinders-group heat sink is given. The investigations show that the tip clearance has a little influence on flow resistance and Nu of micro-cylinders-groups heat sink at low Re and this influence becomes more apparent with the increase of Re . The heat transfer efficiency of micro-cylinders-group heat sink increases rapidly with Re at first, and then becomes slowly, even begins to drop with the further increase of Re . However, the existence of the tip clearance promotes the heat transfer efficiency in micro-cylinders-group heat sink in the present study.

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1. Introduction

Traditional cooling technology can no more satisfy the need of the rapid development of complementary metal oxide semiconductor, so some new cooling technologies have been found to employ in wide applications, such as the cooling technology using micro-scale short pin fin channels [1]. Micro-cylinder-groups (we called the circular fin pins as the cylinders-group) can be used to enhance the convective heat transfer for that the perturbation of micro-cylinders to the flow may advance the transition from laminar to turbulence flow. In the design of high efficient micro-cylinders-group heat exchanger [2], the tip clearance affects the flow and heat transfer characteristics in the micro-cylinders-group, which makes the flow and heat transfer in the micro-cylinders-group more complicated.

Over the last ten years, a number of investigations had been carried out on single-phase heat transfer [3–5], boiling heat transfer [6–8], and the flow characteristics [9–11] of working fluid over a bank of shrouded micropin fins. Kosar et al. [9,11] investigated experimentally the forced convective heat transfer of refrigerant R-123 flowing through a bank of shrouded micropin fins 243 μm long with hydraulic diameter of 99.5 μm . Experimental results showed that a delay in boiling incipience had been observed and a meta-stable single-phase region was apparent at liquid exit temperatures noticeably exceeding the saturation temperature.

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Furthermore, the endwall effects on the heat transfer diminished when $Re > 100$. They found that the experimental Nu were in a reasonable agreement with conventional theoretical results for the Re above 50. The relationships between Nu and Re were similar with those of the conventional scale. However, Re number has more significant influence on Nu in the micropin fins than that in the conventional scale pin fins. Mizunuma et al. [12] experimentally and numerically studied forced convective heat transfer from a micro-finned surface. It was found that the increased flow velocity in the free flow area above the fins in higher height channels yielded high heat transfer coefficients. Avisar et al. [13] investigated experimentally the velocity field in a finite cylinder array in a water towing tank and an acoustic Doppler velocimeter (ADV). The three components of the velocity vector were measured at various points within the model and time-averaged results are presented for the different flow configurations. The results provide insight into the development of the flow field in cases of a finite array with complex geometry and boundary effects. Min et al. [14] numerically investigated the influence of the effect of tip clearance on the cooling performance of microchannel heat sink under the fixed pumping power condition. From the numerical results, the optimized tip clearance is determined and the presence of tip clearance can improve the cooling performance of a micro-channel heat sink when tip clearance is smaller than a channel width. Marques and Kelly [15] investigated heat transfer and pressure drop results for flat parallel plate and curved pin fin microheat exchangers with a staggered arrangement for application to gas turbine blade cooling. It was concluded that the micropin fin heat

Nomenclature

A	whole heat transfer area (m^2)	T	wall temperature ($^{\circ}\text{C}$)
c	constant	T_1	average temperature of upper layer of test section sidewall ($^{\circ}\text{C}$)
C_g	tip clearance (m)	T_2	average temperature of under layer of test section sidewall ($^{\circ}\text{C}$)
c_p	specific heat ($\text{J kg}^{-1} \text{K}^{-1}$)	u	flow velocity (m s^{-1})
d	hydraulic diameter of cylinder (m)	V	volume flow rate ($\text{m}^3 \text{s}^{-1}$)
G	mass flux (kg s^{-1})	U	electric voltage (V)
h	convective heat transfer coefficient ($\text{W m}^{-2} \text{K}^{-1}$)	W	width of test section heat sink (m)
H	height of cylinder (m)		
I	electric current (A)		
K	thermal conductivity coefficient ($\text{W m}^{-1} \text{K}^{-1}$)		
L	length of micro-cylinders-group heat sink (m)		
m_f	fin parameter		
n	number of cylinders		
Nu	Nusselt number		
P	heating power (W)		
ΔP	pressure drop between inlet and outlet of micro-cylinders-group heat sink (Pa)		
Pr	Prandtl number		
Re	Reynolds number		
S_1	distance between bottom and upper layer of sidewall of test section (m)		
S_2	distance between under layer and upper layer of sidewall of test section (m)		
S_L	longitudinal distance of cylinder (m)		
S_T	transverse distance of cylinder (m)		

Greek symbols

f	fin efficiency
κ	heat transfer efficiency
μ	dynamic viscosity ($\text{kg m}^{-1} \text{s}^{-1}$)
ρ	density (kg m^{-3})

Subscripts

f	fluid
i	inlet of test section
loss	heat loss
max	maximum
o	outlet of test section
w	wall

exchanger performance was better than a parallel plate counterpart, and the k - ε turbulence model was the most commonly used model for prediction of thermal and hydrodynamic characteristics of pin-fin micro-heat exchangers. Peles et al. [16] investigated heat transfer and pressure drop phenomena of the water over a bank of micropin fin channels. The paper analyzed the influence of the geometrical and the thermo-hydraulic parameters on the total thermal resistances. Galvis et al. [17] numerically investigated the micro-heat exchanger (MHE) with small staggered cylindrical pin fins with hydraulic diameters of 0.5, 5.1 and 8.5 mm, respectively. The investigation showed that the flat micropin fin heat exchanger overall thermal performance always exceeded that of the parallel plate counterpart (smooth channel) by a factor as much as 2.2 for the 8.5 mm diameter pins, and by 4 for the 0.5 mm diameter pins in the investigated Re range. Furthermore, among the six turbulence models investigated, the RNG model tended to be the best model to predict both the Nu and the friction factor and capture the main feature of the flow field in MHE. Moores and Joshi [18] experimentally studied that the influences of tip clearances on pressure drop performance and heat transfer characteristics. The investigation showed that the influence of tip clearance on pressure drop performance was still apparent even at $Re < 5 \times 10^3$. They also found that heat transfer rates were higher at the dimensionless clearances (C_g/D) < 0.2 than the case with no tip clearance. Besides, it was found that Nu was to be only 1.5% lower at $C_g/H = 0.25$ compared to the case with no tip clearance. Ohadi et al. [19] over-viewed several thermal management techniques that could serve as an effective means to cool the next generation high flux electronics. These included the immersion cooling, jet impingement and spray cooling, and ultra thin film evaporation (UTF), particularly to serve at the chip level for hot spot cooling of the chip. The potential of these techniques for thermal management of current high heat flux electronics as well as the promise they hold for spot cooling of the next generation high flux electronics for commercial and military sectors were also discussed. Naphon and Sookkasem [20] experimentally and numerically investigated the heat transfer characteristics of the in-line and staggered taper

pin fin heat sink under constant heat flux conditions. The tapered pin fin heat sink was fabricated from square aluminum with length, base and tip diameters of 67, 13 and 7.24 mm, respectively. Experiments were performed at various Re numbers in the range of 1000–9000 and heat fluxes in the range of 0.91–3.64 kW/m^2 . The number of pin fins for the in-line and staggered configurations were 16 and 17, respectively. The k - ε standard turbulence model was employed to simulate the turbulent heat transfer characteristics. A finite volume method with an unstructured non-uniform grid system was employed for solving the model. The predicted results were validated by comparing with measured data and were in reasonable agreement with the experiments. Rozati et al. [21] numerically analyzed cylindrical pin fins with tip clearances in the low Reynolds number range ($Re < 400$) in a plane minichannel. They identified a number of unique flow effects, which increase the heat transfer performance. The tip gap affects the heat transfer coefficient by eliminating viscosity dominated end wall effects on the pin, by eliminating the pin wake shadow on the end walls, by inducing accelerated flow in the clearance, by reducing or impeding the development of recirculating wakes, and by redistributing the flow along the height of the channel. Moores et al. [22] experimentally evaluated shrouded pin fin arrays with various tip clearances. Pressure loss was measured ($200 < Re < 2 \times 10^4$) and liquid crystal thermography (LCT) was employed to obtain temperature distributions. With LCT measurement, they observed the introduction of tip clearance, the pin tips experience transient fluctuations in flow pattern with presumed vortices washing across the outer edges of the pin tips, producing an elongated vertical temperature profile along the streamwise direction of the tip. Diez et al. [23] calculated the performance of micropin fins of variable diameter and rough surface by means of an approximate procedure based on truncated power series. The geometric effect of roughness was modeled after Bahrami et al., which greatly increased the intricacy of the fin equation, preventing its analytical solution even by symbolic computer codes. The approximate series solution was developed firstly by estimating an adequate number of terms based on convergence to the exact solution of smooth pin fins of

Table 1
The ratio of the tip clearance C_g and the cylinder height H .

Serial	1	2	3
In line	0	0.043086	0.188094
Serial	4	5	6
Staggered	0	0.08241	0.194097

the same geometry. Then, residual convergence for an increasing number of terms was studied for the rough fin. Three selected geometries (hyperbolic, trapezoidal and concave parabolic) were analyzed. Influence of surface roughness was evaluated for a wide range of heat transfer conditions; results were discussed in terms of the two primary quantities of interest in fin design, viz., efficiency and effectiveness. Due to the easiness of the present methodology, it could be safely applied to other geometric arrangements involving straight and annular fins. Reyes et al. [24] studied the effect of tip clearance on micro-channel flow based thermal control systems when the flow itself cannot be considered as fully developed. The investigations showed that the heat transfer (at the lowest volume flow rate, $Re = 416$) was 83% of the baseline configuration for the optimum tip clearance height. However, the required pumping power was only 18% of the baseline case. Tullius et al. [25] performed an optimization study on micro-structure geometry and maximize heat transfer dissipation through convection from a heated surface. Six pin fin shapes were used in a staggered array and attached to the bottom heated surface of a rectangular minichannel and analyzed. Correlations describing the Nusselt number and the Darcy friction factor were obtained and compared to previous ones from recent studies. These correlations only apply to short fins in the laminar regime (see Table 1).

Although a lot of valuable conclusions and literatures on the flow and heat transfer of the micropin fin have been conducted, these studies mainly focus on the influence of geometrical sizes, the array arrangements and the shapes of micropin fins on the flow and heat transfer characteristics. There are very few literatures to systematically study the influence of the tip clearance on the flow and the forced convective heat transfer in micro-cylinders-group, let along the influence of the tip clearance on the heat transfer efficiency of micro-cylinders-group heat sink. On account of above reasons, more experimental investigations about flow and heat transfer in micro-cylinders-group heat sink with different tip clearances are conducted in this paper.

In the present investigation, an experimental study on the flow and heat transfer performance of de-ionized water flowing through the channel of shrouded in-line and staggered micro-cylinders-group heat sinks with different tip clearances are carried out, and the heat sinks are made up of micro-cylinders with hydraulic diameter of 0.5 mm and heights of 0.5 mm, respectively. As the Re varies from 8 up to 400 in the experiment, the friction factors are measured at different Re and the experimental Nu are also obtained at different Re and with the approximately constant heat flux heating. Based on the experimental results, the heat transfer efficiency in micro-cylinders-group heat sink is systematically analyzed, and explanations are made for the heat transfer enhancement mechanism in micro-cylinders-group heat sinks due to the existence of the tip clearance.

2. Experimental setup and procedure

2.1. Experimental setup

The test rig was schematically illustrated in Fig. 1.

As shown, the apparatus mainly consists of a pressure supply system, a liquid pool, a test section, a data acquisition system

and an IR camera (FLIR, T400), etc. The stable and exact pressure nitrogen is supplied by the pressure supply system, which is formed by a nitrogen bottle, a three-layer filter, a precise pressure regulating valve and a quick-opening valve. The pressure supply system, which is capable of pressure to 12 MPa, maintained precisely a given pressure (less than 1.6 MPa) by a precise pressure regulating valve. In order to avoid any particles in the nitrogen entering and contaminating the working fluid, the three-filter, with the aperture of filtration film of 50 μm , 25 μm and 5 μm in turn, are installed between the nitrogen bottle and the precise pressure regulating valve. The function of gas storage reservoir is used to avoid gas fluctuation.

De-ionized water is used as the working fluid, and it is deposited in the liquid pool. De-ionized water driven from the liquid pool by the high-pressure nitrogen flows through the test section. The liquid pool is connected by a small-diameter polyurethane tube with 2 MPa resisting pressure to a joint just upstream of the test section. Two calibrated pressure transducers, which are installed at the two-end joint of the test section, are used to measure the inlet and the outlet pressures of the test section and the transducers are zeroed before each test. Precise liquid temperature measurements are necessary to determine the accurate values of the viscosity and the density. A T-type thermocouple (Omega Company Product) is placed at the outlet of test section to measure the exit temperature of the flow. Another thermocouple with the same type and accuracy is placed at the inlet to measure the entrance temperature of the flow, as shown in Fig. 1. Considering fluid flowing from the test section will enter the polyurethane tube, in which it is mixed and the temperature is measured by the thermocouple, so there is no need of special arrangement for this measurement in the present experiments. Then the measured temperatures are used to determine the mean fluid temperature and to assess its thermal properties. The fluid flow rate for a given pressure difference and a micro-cylinders-group sample is determined by collecting a liquid volume during a corresponding elapsed measuring time. An electronic balance (FA2004-Type) is used to measure the weight of the accumulated liquid. The measurement is initiated after the prescribed experiment pressure is set by opening the reducing valve in the nitrogen bottle and adjusting the precise regulating valve, which can supply pressure gas of $10\text{--}1.6 \times 10^6$ Pa until the fluid flow becomes steady and the fluid level in the measuring cup reached a pre-set minimum. The elapsed time is then recorded while the level in the measuring cup reached a desired increment.

The IR camera used in the present study has a measuring range from -50 to 2000 $^{\circ}\text{C}$ and the maximal measuring accuracy is 0.1 $^{\circ}\text{C}$. Principally, the surface temperature of an object can be measured by an IR camera if the surface emissivity of the measured object is known. A special glass with zinc sulfide, through which 99.9% of infrared ray can permeate, is used to cover the base plate. Considering the temperature fields of the micro-cylinders are too small to be shot by the IR camera, a special magnifying lens is used in the present experiments. The real surface emissivity of the red copper can be found in common material emissivity tables involved in the IR camera instruction book and it is inputted into the IR camera before the experiments is conducted.

2.2. Test section

Fig. 2 gives a detailed view of the test section.

In order to eliminate contact thermal resistance between the test section and the heating system, an integrative design method is used for precision machining of the test section and the heating part, namely, a copper bar is machined to the test section on the top and the heating system on the bottom, and nine electrical rods are placed inside the bores in the under part of the copper bar, as

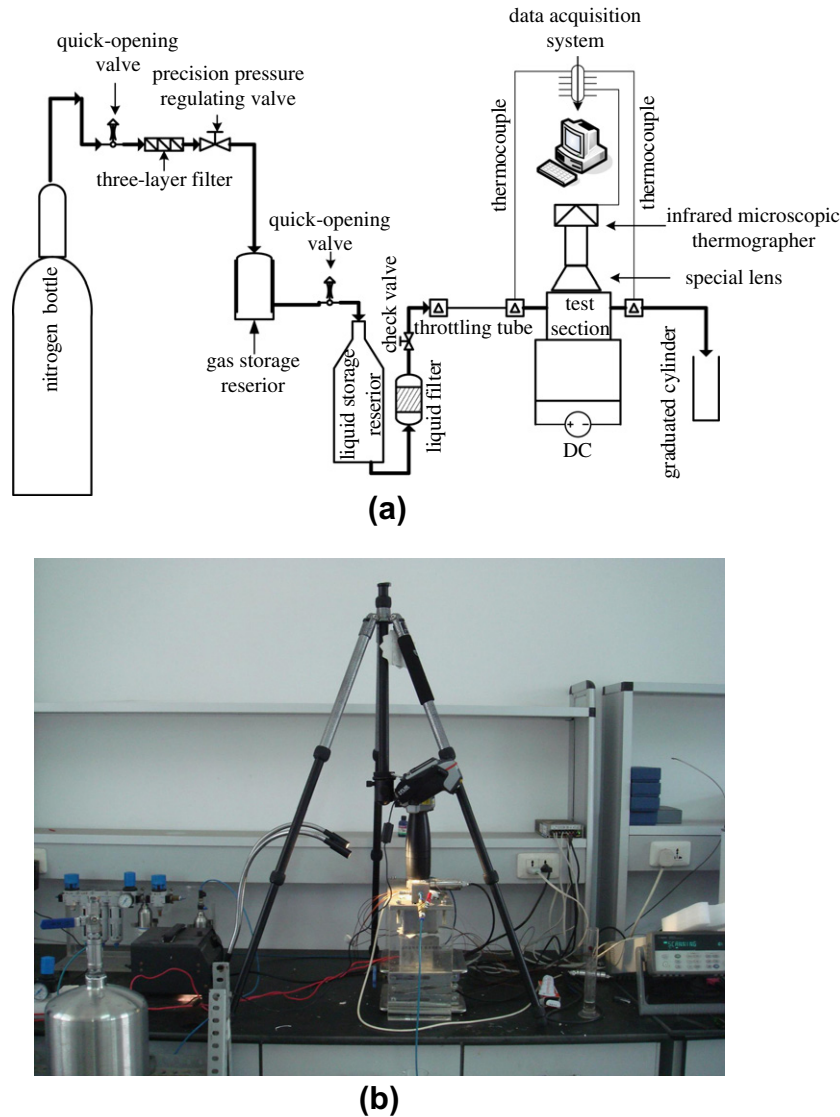


Fig. 1. Experimental loop of micro-cylinders-group heat transfer. (a) Schematic illustration of the experimental loop. (b) Photo of experimental setup.

shown in Fig. 2a and c. Using precision machining, the test section base plates consisting of the in-line and the staggered micro-cylinders-group heat sinks with a 40 mm-long, 5.2 mm-wide are obtained and shown in Fig. 2b. Length L and width W of each base plate are measured by a vernier, while the hydraulic diameter d , the height H of micro-cylinder, the longitudinal distance of cylinder S_L and transverse distance of cylinder S_T are measured by a micrometer. Considering $d = H = S_L = S_T = 0.5$ mm in the present experiments, the flow in micro-cylinders-group heat sink can reach fully developed due to $L/d > 0.05RePr$ for the laminar flow or $L/d > 60$ for the turbulent flow [26]. A special thin glass layer with zinc sulfide covers the base plate and 704-silica gel is used to air-proof the base plate and the thin glass, so the micro-channels made from the micro-cylinders are formed. Due to the existence of an uneven surface of the thin glass and the thickness of the 704-silica gel, the tip clearance is coming into being, as shown in Fig. 3b. The surface irregularity degree of the glass can be measured by a glass flatness measuring instrument (RWG-I /LVDT/DF, Strainoptics Co.) and the thickness of silica gel by a micrometer, so the ratio of the tip clearance C_g dimensions of micro-cylinder base plate and H of micro-cylinder are obtained and listed in 1.

The six K-type thermocouples (Omega Company Product) are placed in the joint between the test section and heating part, which

are arrayed at the inlet, the medium and the outlet of the test section (upper layer and under layer) along flowing direction of the working fluid to measure the average temperatures of the upper layer and under layer of the test section, as shown in Fig. 2a. A DC stabilized source (SDC36100S) is used to supply electrical heating power and an amperemeter and a voltmeter are connected to the heating system and they are used to measure the heating power.

2.3. Experimental uncertainties

In the present experiment, the main uncertainties come from the measurements of the temperature, the pressure drop, the flow flux, the heating power and the geometrical dimensions of micro-cylinders-group heat sink. The two T-type thermocouples with accuracy of 0.15°C (less than 100°C) are used to measure the inlet and the outlet temperature of the test section and the six K-type thermocouples with accuracy of 0.25°C (less than 400°C) measuring outer surface temperature of the test section; the minimal temperature difference between the inlet and the outlet of staggered array sections within the range of the present experiment is about 10.7°C at $Re = 397.2$, and thus the maximal relative temperature uncertainty in evaluating the heat transfer rate is about 2.81%;

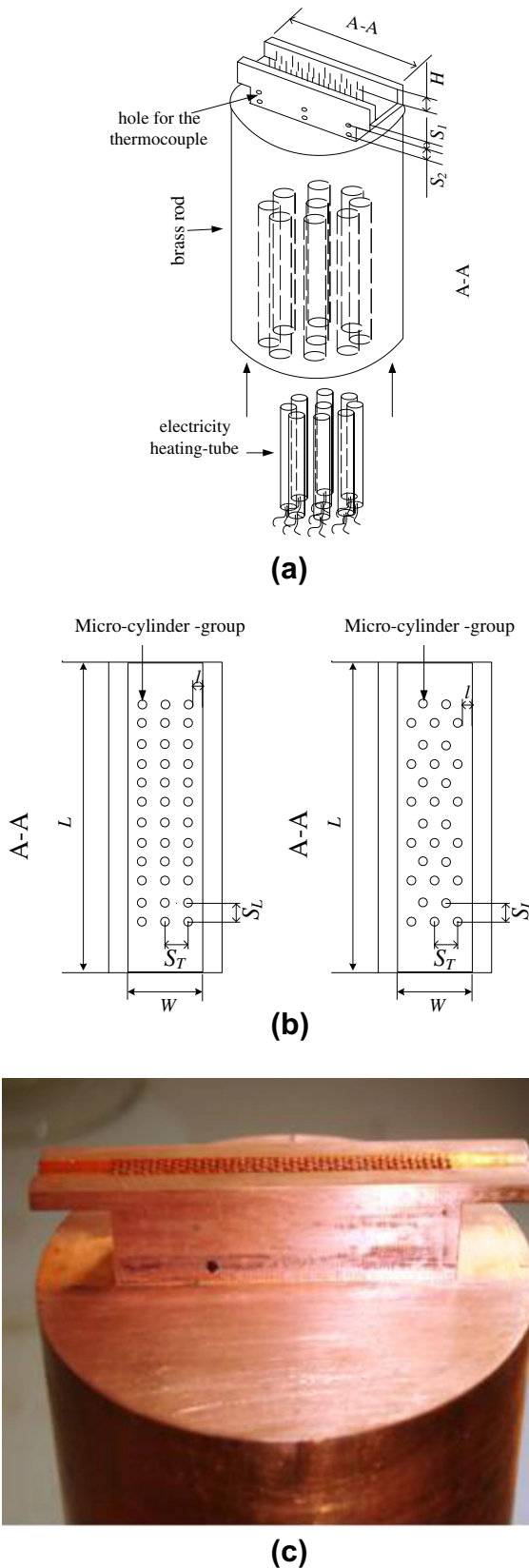


Fig. 2. Test section of micro-cylinders-group. (a) Heating system in-line array staggered array. (b) Schematic of micro-cylinders-group test section. (c) Photo of the test section.

the uncertainty of the pressure is less than ± 1 kPa for the pressure transducers of CYB-10S type; an electronic balance (FA2004-Type)

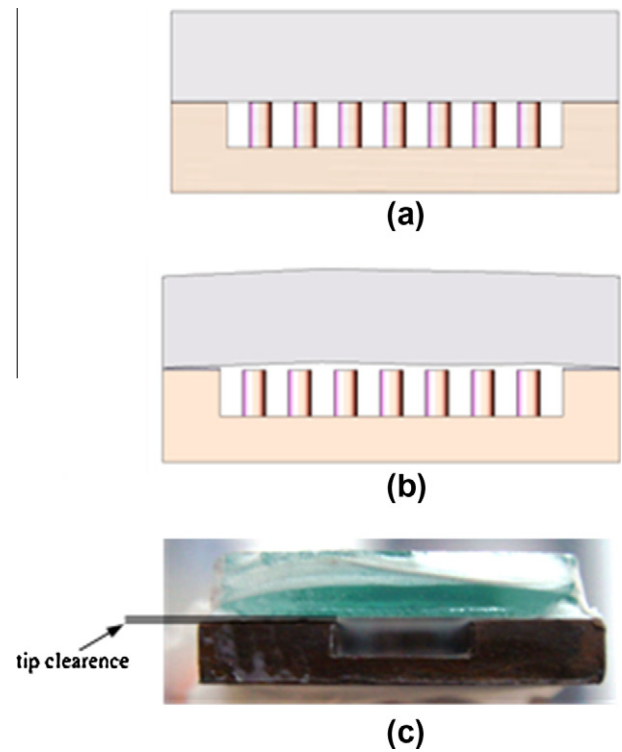


Fig. 3. Schematic of test section with and without tip clearance. (a) Test section without tip clearance. (b) Test section with tip clearance. (c) Photo of the test section with tip clearance.

Table 2
Experimental uncertainties.

Parameter	Uncertainties
Temperature difference, ΔT	± 0.3 °C
Flow rate, V	$\pm 1.1\%$
Electric power, P	$\pm 0.03\%$
Wall temperature, T	± 0.6 °C
Reynolds number, Re	$\pm 4.2\%$
Friction factor, f	$\pm 5.84\%$
Nusselt number, Nu	$\pm 6.5\%$

with an accuracy of ± 0.1 mg is used to measure the weight of the accumulated liquid, so the volume of accumulated liquid can be calculated considering the density of the working fluid; an ammeter and a voltmeter, whose accuracies are 0.02% and 0.01% of the maximum measurement range, respectively, are used to measuring the current and the voltage of electric source. The uncertainties of geometrical dimensions of micro-cylinders-group heat sink are ± 5 μ m which are dependent on the accuracy of the engraving machine (YF-DA7060), so the relative roughness of the bottom and the cylinder surface is less than 2%.

A stable state is achieved when the pressure and the temperature keep a constant level, and the experimental data can be collected in this state. In the present experiment, all other data can be acquired from the data acquisition system except for the volumetric flow rate. As mentioned above, the experimental uncertainties of all data are listed in Table 2, and the error analysis method comes from reference literature [27].

3. Results and discussion

As shown in Fig. 2c, the average temperature of the contact bottom surface of the test section is as follows based on the hypothesis of one-dimensional steady heat conduction:

$$T = T_1 - \frac{(T_2 - T_1) \times S_1}{S_2} \quad (1)$$

where T_1 and T_2 are the average temperatures of upper layer and under layer of test section, °C, respectively, which are measured by the six K-type thermocouples, as shown in Fig. 2a.

The inlet and the outlet temperatures, T_i and T_o , respectively, are measured during the experiments process. The average temperature of the inlet and the outlet is used as the characteristic temperature and the hydraulic diameter of micro-cylinder as the characteristic dimension, so the Re is expressed as:

$$Re = \frac{\rho u_{\max} d}{\mu} \quad (2)$$

$$u_{\max} = \frac{GS_T}{\rho WH(S_T - d)} \quad (3)$$

In the present experiments, the measured parameters are the pressure drop across the micro-cylinder group ΔP , the volume flow rate V and the temperature of the working fluid T_f . Other parameters are used to describe the friction characteristics in micro-cylinders-group heat sink. The friction factor f can be easily related to these parameters as follows:

$$f = \frac{2\Delta P d}{\rho Lu_{\max}^2} \quad (4)$$

When the working fluid flows through the micro-cylinders-group heat sink, the area of whole heat transfer surface can be expressed:

$$A = \eta_f n \pi d H + 2HL + WL - \frac{n\pi d^2}{4} \quad (5)$$

The convective heat transfer coefficient in the micro-cylinders-group heat sink can be obtained based on the cooling law of Newton:

$$h = \frac{(P - Q_{\text{loss}})}{\left(\eta_f n \pi d H + 2HL + WL - \frac{n\pi d^2}{4}\right) \left(T - \frac{T_i + T_o}{2}\right)} \quad (6)$$

$$P - Q_{\text{loss}} = c_p G(T_o - T_i) \quad (7)$$

$$\eta_f = \frac{th(m_f H)}{m_f H} \quad (8)$$

$$m_f = \sqrt{\frac{h}{k_w d}} \quad (9)$$

$$Nu = \frac{hd}{k_f} \quad (10)$$

The working fluid flows vertically into the test section at the middle. In present study, the IR-technique is used to measure the temperature distribution of the test section, which judges the flow field changes. Fig. 4 shows the temperature field on micro-cylinders-group heat sink surface at same heating power and different Re can be obtained by the IR camera with a special infrared magnifying lens. Water temperature is about 23.4 °C and the flow direction is left-to-right in Fig. 4.

Fig. 4 shows the surface temperature distribution of the no. 6 test section at different Re . The working fluid cannot saturate the whole channels in the micro-cylinders-group heat sink when it flows through test section at extremely low Re . Meanwhile, there is no working fluid between the top of micro-cylinders and the packaging glass (namely, tip clearance), because the surface temperature field can be distinctly shot by the IR camera, as illustrated

in Fig. 4a. Before the working fluid soaked into the packing glass, there is no obviously influence of the tip clearance on the flow and heat transfer in micro-cylinders-group heat sink. With the increase of Re , the working fluid has gradually filled in all flow channels of micro-cylinders-group, so part of working fluid flows through the tip clearance so that part regions of packaging glass are soaked by the working fluid at $Re = 45.7$ and $Re = 107.8$, therefore, the temperature distribution image is a little vague, as shown in Fig. 4b and c. The working fluid evenly flows through whole tip clearance of the test section at $Re > 199.9$ and the packaging glass is completely soaked by the working fluid with the further increase of Re . Therefore, the surface temperature field images become distinct, as shown in Fig. 4d–f, and the tip clearance may affect the flow and heat transfer characteristics in micro-cylinders-group heat sink.

As shown in Fig. 4, the temperature difference is very small at some regions according to the analysis of the temperature distribution, so there are small or no temperature gradients along the flow direction at some regions of micro-cylinders-group surface at low Re , even the temperatures of micro-cylinders are less than those of the bottom of the baseplate at extremely low Re , as shown in Fig. 4a, which shows no working fluid flows through those regions or the flow stagnates at those regions, and thus the sluggish regions are formed. The temperature gradients obviously increase along flow directions at relatively high Re , and the areas of the sluggish regions gradually diminish. As a result, the average temperature of test section apparently drops, shown in Fig. 4c and d. The sluggish regions at the cylinders-group region will disappear with further increase of Re except for the regions between outermost cylinders and the wall of micro-cylinders-group plate, shown in Fig. 4e and f, which is resulted from the flow stagnation due to the viscosity of working fluid.

The variations of the experimental friction factor with Re are plotted in Fig. 5 in micro-cylinder-group heat sinks at in-line and staggered array arrangements for the same distances among cylinders.

According to Fig. 5, there is little influence of the tip clearance on the flow resistance at extremely low Re , but the tip clearance affects the flow resistance in micro-cylinders-group heat sink when $Re > 50$ and the flow resistance of micro-cylinders-group with the tip clearance is less than that of the non-clearance case. Part of water flows through the tip clearance during the course of the experiments, which can apparently reduce the flow resistance, especially for the higher value of C_g . Sparrow and Ramsey [28] employed a duct of constant height with pins of varying length resulting in H/D from 1.75–2.5 and C_g/H of 0, 0.11, 0.25 and 0.43. For C_g/H of 0.11, the investigations found f was reduced to 41% of the non-clearance case. In the present study, the experimental data of $C_g/H = 0.086$ and $C_g/H = 0.188$ for the in-line arrays are about 24.1% and 44.3% lower than those of the non-clearance case at $Re = 395$ and the difference between the clearance case and the non-clearance case is becoming greater and greater with the further increase of Re . For the staggered arrays, the experimental friction factor also decreases with the increase of C_g for $Re > 50$. The difference among the test sections for different values of C_g is becoming larger and larger with the increase of Re and the maximal difference between $C_g/H = 0$ and $C_g/H = 0.194$ reaches 56.2% based on the experimental results.

In addition, the difference of the friction factor between $C_g/H = 0$ and $C_g/H = 0.086$ for the in-line array is less than that between $C_g/H = 0$ and $C_g/H = 0.082$ for the staggered array. However, the difference between $C_g/H = 0.086$ and $C_g/H = 0.188$ for the in-line array is nearly close to that between $C_g/H = 0.082$ and $C_g/H = 0.194$ for the staggered array. It is well known that the flow resistance for the staggered array is much larger than that for the in-line array at relatively high Re [29], so the friction factor difference between in-

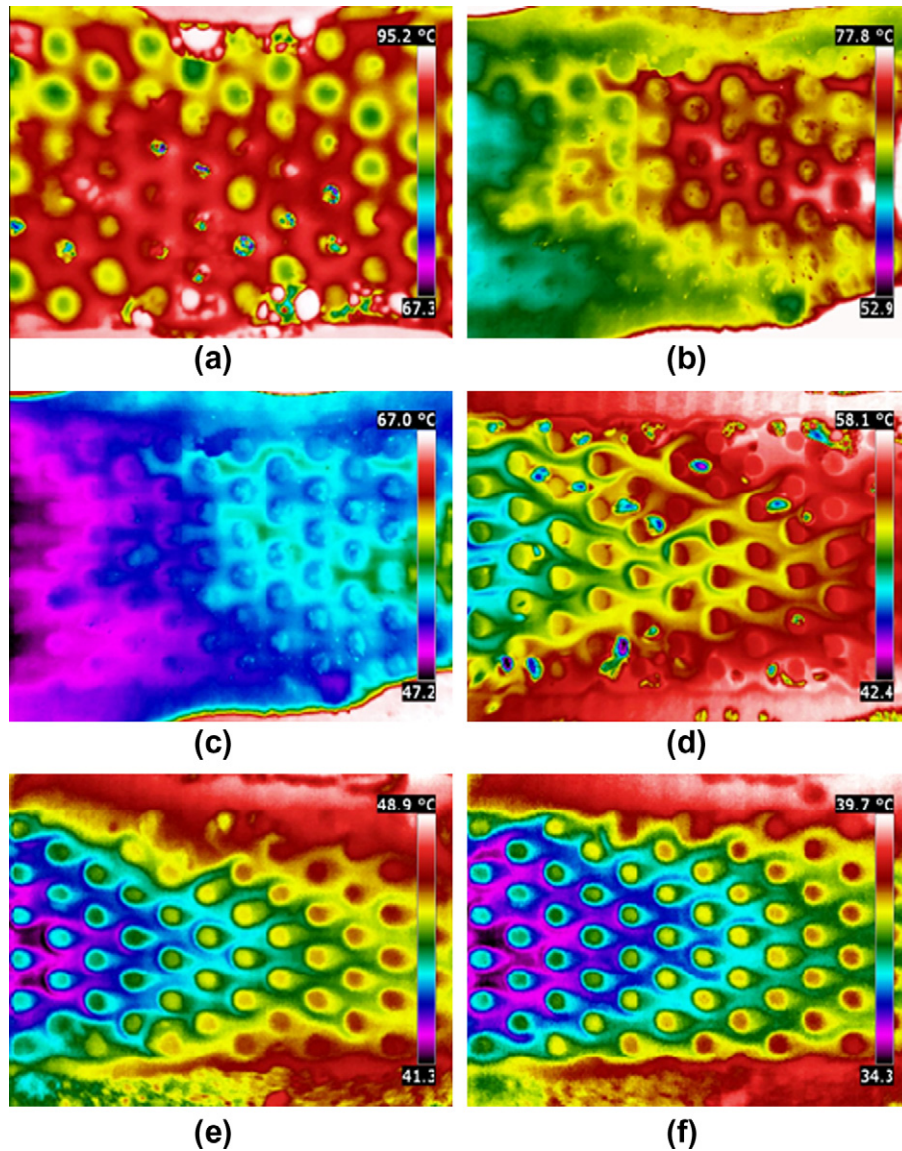


Fig. 4. IR image of micro-cylinders-group heat sink surface (the no. 6 test section). (a) $Re = 15.2$. (b) $Re = 45.7$. (c) $Re = 107.8$. (d) $Re = 199.9$. (e) $Re = 311.9$. (f) $Re = 394.1$.

line micro-cylinders-group and the test section with tip clearance of $C_g/H = 0.086$ is less than that between staggered micro-cylinders-group and test section with $C_g/H = 0.082$. At the same Re ($Re > 50$), more fluids will flow through the tip clearance for the staggered micro-cylinders-group heat sink with $C_g/H = 0.082$ than the in-line one with $C_g/H = 0.086$, so the pressure drop for the staggered micro-cylinders-group heat sink will larger than that of the in-line array one. With the increase of the tip clearance, the fluid flow rate through the tip clearance is still larger for the staggered array compared with the in-line array at the same Re , but the ratio of friction factor difference between $C_g/H = 0.086$ and $C_g/H = 0.188$ to friction factor with $C_g/H = 0$ at the in-line array is larger than that of the friction factor difference between $C_g/H = 0.082$ and $C_g/H = 0.194$ to friction factor with $C_g = 0$ at the staggered array at the same Re based on the experimental results analysis.

The temperatures of the inlet and the outlet of test section are measured by the thermocouples, and the experimental Nu can be measured based on Eqs. (2)–(9). The relationships between the experimental Nu and Re are shown in Fig. 6.

Fig. 6 shows the experimental Nu increasing with the increase of Re as expected. At low Re , especially at $Re < 50$, the sluggish

regions and the endwall effect obviously drop the experimental Nu , which is the same as the results of literature [12]. Literature [12] thought the above experimental phenomenon was due to the existence of the endwall effect. However, there is the appearance of the sluggish regions in the micro-cylinders group heat sink at low Re , which also results in the dramatic decrease of the experimental Nu besides the endwall effect, as illustrated in Fig. 4. There is no much difference about the experimental Nu among different tip clearances at $Re < 100$ according to Fig. 6. With the increase of Re and the tip clearances, the difference of the experimental Nu will be more apparent, and the maximal differences between $C_g/H = 0$ and $C_g/H = 0.188$, $C_g/H = 0$ and $C_g/H = 0.194$ for the in-line array and the staggered array are 15.4% and 22.7%, respectively, based on the present experiments.

For $C_g = 0$, there is no gap between the top of micro-cylinders and airtight shell (which is the glass in the present experiments) and the heat emission from the top of micro-cylinders by heat conduction is relatively small, so the axial temperature gradient of the micro-cylinders is small at the steady working state, namely, the heat releasing by the axial heat conduction of micro-cylinders is also very small. For $C_g \neq 0$, part of water flow through the gap

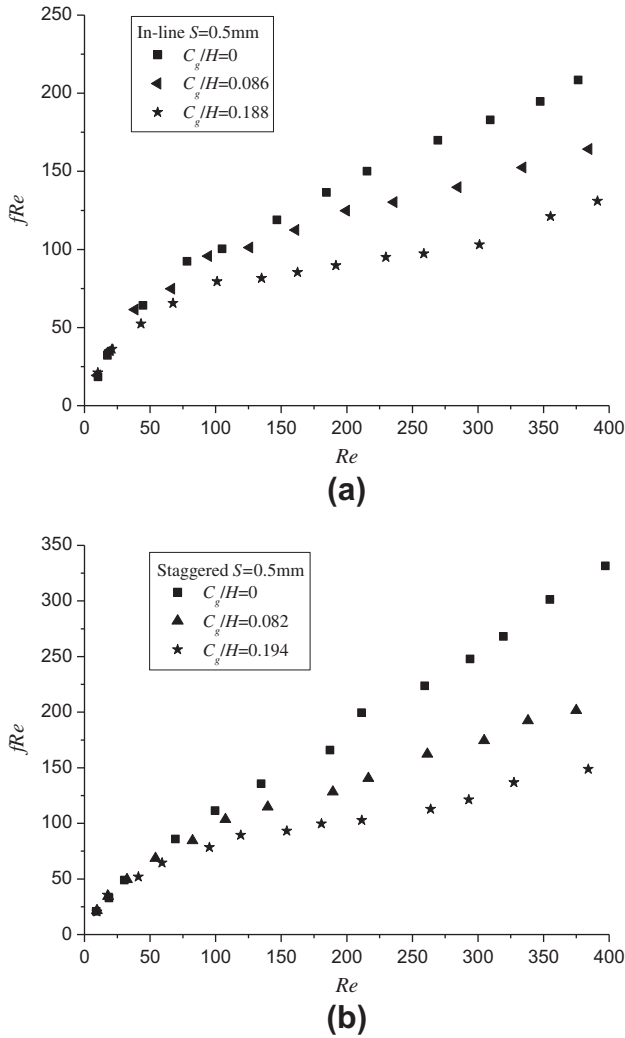


Fig. 5. Influence of tip clearance on flow resistance of micro-cylinders-group heat sink: (a) In-line array. (b) Staggered array.

between the top of micro-cylinders and the glass (namely, the tip clearance), so the heat quantity from the top of micro-cylinders is emitted by the forced convective heat transfer, so the axial temperature gradients of the micro-cylinders are larger compared with those of $C_g = 0$. On one hand, there are some following important effects for $C_g \neq 0$ [21]: (1) the clearance eliminates viscosity dominated endwall effects on the cylinder; (2) it eliminates the cylinder wake shadow on the endwall with the clearance; (3) it induces accelerated flow in the clearance, which affects the endwall and the cylinder-top surfaces; (4) it reduces or impedes the development of recirculating wakes by enhanced mixing of the tip flow and the wake; and (5) it redistributes the flow along the height of the channel. These effects carry different relative imports at different Re and contribute to increasing the heat transfer coefficient; on the other hand, there are higher temperature gradients for $C_g \neq 0$ from the bottom to the top of micro-cylinders compared with those of $C_g = 0$ and the heat conduction quantity by the axial micro-cylinders and the convective heat transfer quantity by the top of micro-cylinders rises significantly for $C_g \neq 0$. In addition, it is easier for micro-cylinders-group to suffer from the disturbance and give rise to vortexes and an earlier transition from laminar flow to turbulent flow at the wake zone though there is more mixing in heat transfer in the tip clearance for $C_g \neq 0$. Therefore, there is the higher Nu in micro-cylinders-group than in micro-channel

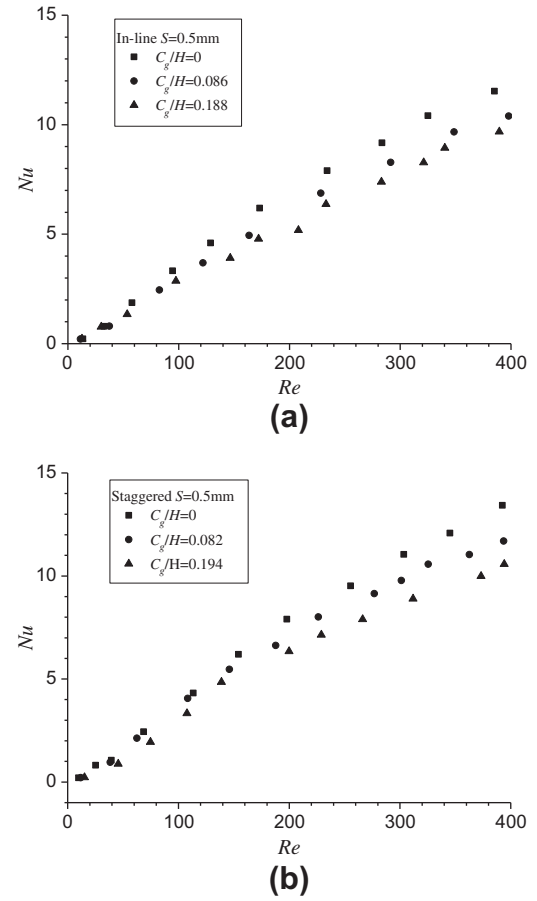


Fig. 6. Relationships between experimental Nusselt numbers and Reynolds number at different tip clearances. (a) In-line array. (b) Staggered array.

for the same hydraulic diameter [12,30], so part of water that should have flowed by the micro-cylinders-group now flows through the tip clearance, as a result, the experimental Nu is reduced. Moreover, the temperature difference between the fluid and micro-cylinder wall is decreased due to the higher temperature gradient on the micro-cylinder wall at $C_g \neq 0$, which further weakens the heat transfer for $C_g \neq 0$, as shown in Fig. 6.

To sum up the above analyses, the heat transfer in micro-cylinders-group heat sink with tip clearance is weakened but the difference of Nu between test section with tip clearance and without tip clearance is relatively small, as illustrated in Fig. 6. The non-dimensional parameter, κ , is heat transfer efficiency and it can be expressed:

$$\kappa = \frac{Nu}{fRe} \quad (11)$$

The κ shows transfer as many heat quantity as possible used in a certain amount of mechanical energy, namely, the higher the heat transfer efficiency is, the larger the κ is. The variations of the κ values for different tip clearances with Re are plotted in Fig. 7 at the in-line and the staggered arrays.

Whether at the in-line array or at the staggered array, the κ value rapidly increases with Re for a start according to Fig. 7. With the increase of Re , the κ value slowly increases, and then begins to drop when the Re exceeds a certain value. Moreover, the longer the tip clearance is, the larger the κ value becomes based on present experimental results, and this is because that the decrease ratio of friction factor is much larger than that of the experimental Nu , as described above.

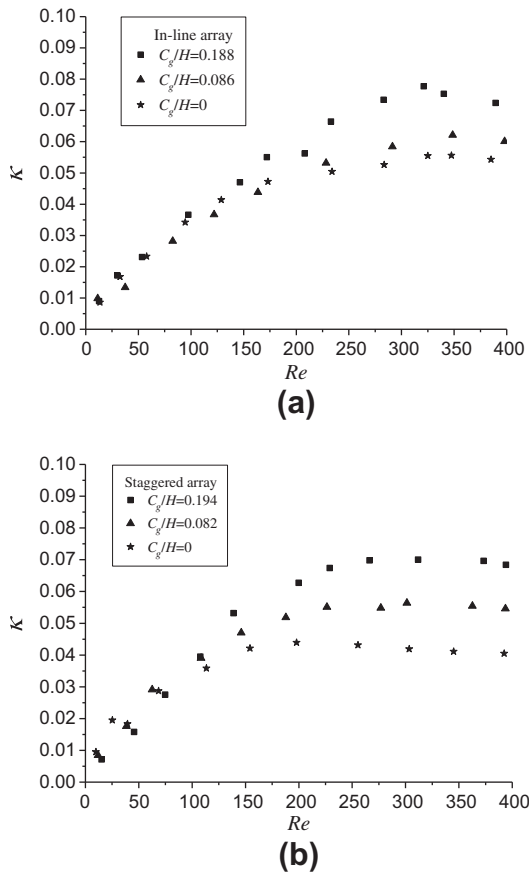


Fig. 7. Comparison of the κ values for different tip clearances with Reynolds number. (a) In-array. (b) Staggered array.

For the in-line array, the κ value differences among tip clearances are relatively small at $Re < 150$ and then become larger with the increase of Re . The present experimental results show the maximal κ value difference between $C_g/H = 0.188$ and $C_g/H = 0.086$ is about 32.4% and the κ value for $C_g/H = 0.086$ is about 8.67% higher than that for $C_g/H = 0$, which shows the micro-cylinder group heat sink with relatively longer tip clearance will be beneficial to heat transfer enhancement.

For the staggered array, the variation trend of the κ value is the same as that of the in-line array, but the κ value differences among the different tip clearances become more apparent compared with those of the in-line array, as illustrated in Fig. 7. Deviating from the variation law for κ value of the in-line array, the κ value difference between the $C_g/H = 0.194$ and the $C_g/H = 0.082$ is close to that between the $C_g/H = 0.082$ and the $C_g/H = 0$, and the maximal difference between the $C_g/H = 0.194$ and the $C_g/H = 0.082$, the $C_g/H = 0.082$ and the $C_g/H = 0$ is 28.9% and 41.3%, respectively, so the less tip clearance for the staggered array will lead to more apparently heat transfer enhancement.

Stated thus, the tip clearance will enhance heat transfer in micro-cylinders-group heat sink according to the theory of heat transfer enhancement though it drops the forced convective heat transfer coefficient in micro-cylinder-group according to present experimental results.

4. Conclusions

De-ionized water flows through in-line and staggered array micro-cylinders-group heat sink with different tip clearances and the influence of the tip clearance on heat transfer efficiency of micro-

cylinders-group heat sink is experimentally studied. Several conclusions obtained by the investigation are as follows:

1. The surface temperature distribution of micro-cylinders-group heat sink at different Re can be obtained by IR camera with a special infrared magnifying lens and the sluggish regions are found at low Re . In addition, the working fluid, which flows through the tip clearance, is also observed by the IR images.
2. The existence of the tip clearance drops the flow resistance of micro-cylinders-group heat sink at relatively high Re , which declines the flow resistance at in-line array and at staggered array of heat sinks at maximal extent of 44.3% and 56.2%, respectively, in the present study.
3. Though the experimental Nu in micro-cylinders-group heat sink is decreased due to the existence of the tip clearance, the heat conduction quantity by the axial micro-cylinders and the convective heat transfer quantity by the top of micro-cylinders are the larger significantly for $C_g \neq 0$ than for $C_g = 0$, therefore, the difference of Nu between test sections with tip clearance and without tip clearance is relatively small.
4. The tip clearance will enhance heat transfer in micro-cylinders-group heat sink according to the theory of heat transfer enhancement due to the larger decrease ratio for the friction faction than that for the experimental Nu with the increase of the tip clearance.

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